

Assignment: The "Neural Net Navigator"

Objective: Create a visual, hand-written, letter-sized, one-page (double-sided) "Navigator" that guides a fellow student through the logic of manual neural network calculations. Graph paper is the recommended medium.

Your guide should cover the vocabulary and step through the execution of simple Logic Gate (Single Perceptron) as well the workings of complex Classifier (Multi-Layer Perceptron) from input through output.

In the following class, a peer student will only have your guide as a “cheat sheet” to answer questions about or demonstrate the calculations of weights, bias, relu or sigmoid functions in a neural net. You should cover how the formal math notations can be translated into concrete math manipulations (The Tom Yeh’s “by hand” methods would be a good model).

Flag possible “gotchas” or tricky parts that they might have a hard time with because you had a hard time with the concept or task.

Grading:

If the person using your guide gets an “A” on the stress test, you will also receive an “A” as their navigator. If the person using your guide doesn't get an “A”, you will receive a combination of your own points on the test and a peer assessed grade of your guide.

The criteria that the navigation guide would be graded on include: Visual Clarity & Design, Communication, Conceptual Clarity, Comprehensiveness, Scaffolding of concepts.

Examples of questions that could be asked:

Given values for 2 instances with 4 features, preset weights and bias, and relu activation, calculate the outputs of the first hidden layer of this neural network.

Input Layer

	X1	X2
	2	-1
	1	1
	2	1
	1	2

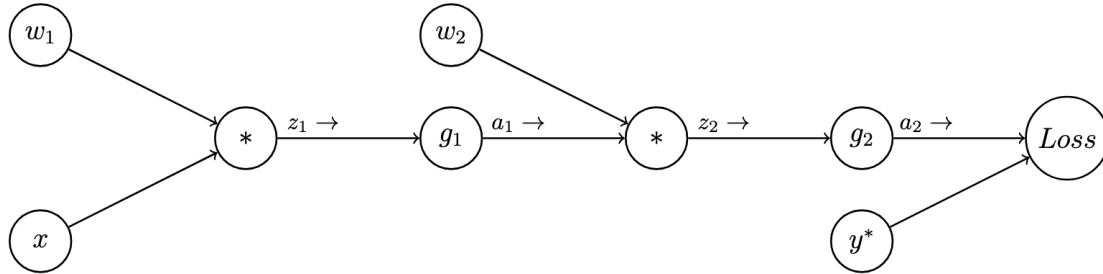
Hidden Layer 1

W1					b1			
1	0	-1	1	1	→	↓	↓	ReLU
0	1	1	0	1	→			~
1	-1	0	0	0	→			~

1 1

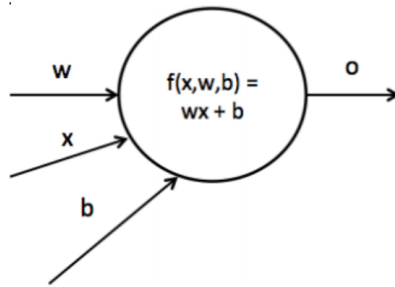
2 Neural Nets

Consider the following computation graph for a simple neural network for binary classification. Here x is a single real-valued input feature with an associated class y^* (0 or 1). There are two weight parameters w_1 and w_2 , and non-linearity functions g_1 and g_2 (to be defined later, below). The network will output a value a_2 between 0 and 1, representing the probability of being in class 1. We will be using a loss function $Loss$ (to be defined later, below), to compare the prediction a_2 with the true class y^* .



1. Perform the forward pass on this network, writing the output values for each node z_1, a_1, z_2 and a_2 in terms of the node's input values:

2. Now we introduce a bias term b into the graph, such that $o = wx + b$ (this is known as an *affine* function). Which of the functions can be represented by this network? For the options which can, write out an appropriate value of w, b .



Sample Cheat Sheet Rubric:

Criteria	Highly Effective (4pts)	Effective (3pts)	Nominally useful (2pts)	Requires Revision (1pt)
Visual Clarity & Design	Layout is professional, inviting, and uses space effectively. Information is dense but not overwhelming. A peer could study from this immediately.	Layout is organized. Text is legible. Good use of the page.	Layout is chaotic or messy. Handwriting or formatting makes it difficult to read.	Disorganized, incomplete, or illegible.
Communication	Language is clear, relevant, actionable, and concise	Language is generally clear. Most labels and notes are helpful, though some might be wordy or vague.	Language is often vague or assumes too much prior knowledge. Hard to understand without help.	Spelling or grammar errors obscure understanding. Language is misleading or lacks relevance
Conceptual Clarity	Uses highly effective visual metaphors (blocks, flows, color-coding) that make the math intuitive without relying solely on formulas.	Uses standard diagrams clearly with good labeling. Mathematical flow is easy to follow.	Relies heavily on abstract equations. Diagrams are cluttered or hard to trace.	Visuals are missing, confusing, or misleading.
Comprehensiveness	Seamlessly covers all topics needed for questions. Explicitly connects the dots between them	The relevant content is available but explanations are merely adequate	Coverage of the material is thin or too brief	Only covers one topic or misses key components
Scaffolding of concepts	Expertly guides the reader from "Simple" to "Complex." Anticipates student confusion.	Progression is logical between the "easy" and "difficult" concepts, but the "why" that links the concepts may be brief.	Hard to see how presented concepts relate to each other. Makes assumptions about user's knowledge	Concepts are presented in random order.
Meta-Learning Awareness	Includes a specific, insights difficulties (e.g., "Where I Got Stuck" section) that identifies conceptual hurdles and actionable tips to avoid them.	Includes a helpful "Tips" or "Common Errors" section. Useful for catching mechanical mistakes (signs, arithmetic).	"Tips" section is vague (e.g., "Check your math") or missing. No real insight into the author's learning process.	No reflection, tips, section included. No signal of meta-learning has been considered